

Wang Xinyi

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Education

- The Chinese University of Hong Kong** 08/2019-Present
Ph.D. Mechanical and Automation Engineering
- **Supervisor:** Prof. Ben M. Chen
 - **Co-supervisor:** Prof. Jie Huang
 - **Research interests:** Task and motion planning, Multi-agent Cooperation System, Trajectory optimization
- Xiamen University** 09/2015-06/2019
B.Eng. Flight Vehicle Design and Engineering
- **GPA:** 91.8/100 (Ranking 1st out of 46)
 - **IELTS:** 6.5 (R6.5, L6.5, S6.0, W6.0) 02/2018

Research Projects

- Development of an Integrated Autonomous Building Inspection and Information Management System** 08/2022-Present
- Design a safety-guaranteed coverage path planning algorithm and an obstacle-aware trajectory generation method for inspection and data collection using.
 - Design a dynamic Voronoi deployment method to reduce total inspection time by distributing a more balanced workload for a multi-agent system, while simultaneously guaranteeing collision avoidance.
- Collaborative Search and Pursuit-evasion for Unmanned Systems in Cluttered Environments** 06/2021-08/2022
- Design a distributed algorithm based on Voronoi partition for cooperatively pursuing an adversarial evader in a forest.
- Advanced Motion Planning Techniques for the Cooperation of Multi-agent Systems** 09/2020-05/2021
- Design a decentralized gradient-free trajectory generation method for multiple quadrotors in dynamic and obstacle-dense environments to achieve distributed and asynchronous cooperative motion planning.
- Intelligent Navigation and Robust Flight Control Systems for Unmanned Systems** 02/2020-12/2020
- Design a guided time-optimal model predictive control based framework to generate replanning trajectories for the quadrotor to track an escaping target in complex environments with consideration of line of sight.
- Development of a Flight Control System for Fixed-wing UAVs** 11/2017-05/2019
- Development L1 adaptive controller and PID controller and applied them on a fixed-wing UAV to track a given flight route.

Publications

(# = equal contribution, * = corresponding author)

- [1] C. Gao, **X. Wang***, X. Chen, B. M. Chen. A Hierarchical Multi-UAV Cooperative Framework for Infrastructure Inspection and Reconstruction. Control Theory and Technology (CTT), 2023, submitted.
- [2] **X. Wang**, Y. Ding, Y. Chen, R. Han, L. Xi*, B. M. Chen. OA-ECBVC: A Cooperative Collision-free

Encirclement and Capture Approach in Cluttered Environments. IEEE Conference on Decision and Control (CDC), 2023.

[3] **X. Wang***, C. Gao, X. Chen, B. M. Chen. Fast and Secure Distributed Multi-agent Coverage Control with An Application to Infrastructure Inspection and Reconstruction. Chinese Control Conference (CCC), 2023, Guan Zhao-Zhi Award Finalist.

[4] **X. Wang**, L. Xi, Y. Ding*, B. M. Chen. Distributed Encirclement and Capture of Multiple Pursuers with Collision Avoidance. IEEE Transactions on Industrial Electronics (TIE), 2023.

[5] Y. Chen*, **X. Wang**, R. Wang, Z. Guo, S. Lai, B. M. Chen. An Interactive System for Multiple-task Linear Temporal Logic Path Planning. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.

[6] Y. Chen, R. Wang, **X. Wang***, B. M. Chen. Sampling-based path planning under temporal logic constraints with real-time adaptation. IEEE International Conference on Robotics and Automation (ICRA), 2023.

[7] C. Gao, **X. Wang**, R. Wang, Z. Zhao, Y. Zhai, X. Chen*, B. M. Chen. A UAV-based explore-then-exploit system for autonomous indoor facility inspection and scene reconstruction. Automation in Construction (AIC), 2022.

[8] L. Xi#, **X. Wang#**, Ding Y, Y. Wei, Z. Peng*, B. M. Chen. Quadrotor trajectory planning for visibility-aware target following. IEEE International Conference on Robotics and Biomimetics (ROBIO), 2021.

[9] **X. Wang**, L. Xi, Y. Chen, S. Lai*, F. Lin, B. M. Chen. Decentralized MPC-based trajectory generation for multiple quadrotors in cluttered environments. Guidance, Navigation and Control (GNC), 2021, 1(02): 2150007.

[10] L. Xi, **X. Wang**, L. Jiao, S. Lai, Z. Peng*, B. M. Chen. GTO-MPC-Based Target Chasing Using a Quadrotor in Cluttered Environments. IEEE Transactions on Industrial Electronics (TIE), 2021, 69(6): 6026-6035.

[11] **X. Wang***, L. Xi, Y. Chen, S. Lai, F. Lin, B. M. Chen. Decentralized trajectory generation technique for multiple unmanned multi-copter systems in cluttered environments. The 12th International Micro Air Vehicle Conference (IMAV), 2021.

Internships

Peng Cheng Laboratory & Shenzhen Weixi Technology Co., Ltd

05/2020-09/2020

Construct Navigation System of Four-wheel Skid-Steering Vehicle in a Park Environment with Pedestrians

- Construct MPC-based vehicle local planner with a trajectory tracking controller to solve problems in tracking the velocity command.
- Test real-time lane line path following and point to point delivery mission using lidar module with considering the dynamic obstacles (with people walking around the robot) and physical limitation of UGV.

Technical Skills

- Experienced in C/C++, Python, MATLAB, LaTeX
- Robotics Frameworks: Robot Operating System (ROS), Gazebo

Teaching Experience

Teaching Assistant:

- MAEG 3080 Fundamentals of Machine Intelligence, instructed by Professor Juan Luis.
- ENGG 5402 - Advanced Robotics, instructed by Professor Fei Chen

Awards and Honors

Third Prize in RoboMaster 2017 (The 16th National College Students Robotics Contest)

08/2017

Xiamen International Bank Scholarship, XMU (Top 4%)

04/2017

Wang Xinyi	Fall 2021
Merit Student, XMU (Twice, Top 8%)	12/2016&12/2017
Outstanding Graduates, XMU (Top 2%)	06/2019
Research Postgraduate Studentship, CUHK	08/2020
Professor Charles K. Ko Student Creativity Award, CUHK	06/2023

Academic services

Reviewer of Journals and Conferences

- IEEE Transactions on Automatic Control
- IEEE Transactions on Cybernetics
- IEEE Robotics and Automation Letters (RA-L)
- SCIENCE CHINA Information Science
- International Journal of Robust and Nonlinear Control
- Autonomous Intelligent Systems
- Guidance, Navigation and Control

- IEEE International Conference on Robotics and Automation (ICRA 2023)
- IEEE Conference on Decision and Control (CDC 2023)
- IEEE International Conference on Control and Automation (ICCA 2022)
- IEEE International Conference on Unmanned Aircraft Systems (IEEE ICUAS 2022)
- Chinese Control Conference (CCC 2023)